

HDOC DAY-9 REPORT

Menu-Driven C Programming Activities

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Batch	15
Activity	HDOC Day-9

Question 1: Swap Two Numbers

Problem Statement

Write a menu-driven C program to input two numbers and swap them using a third variable or without using a third variable. Display the values before and after swapping.

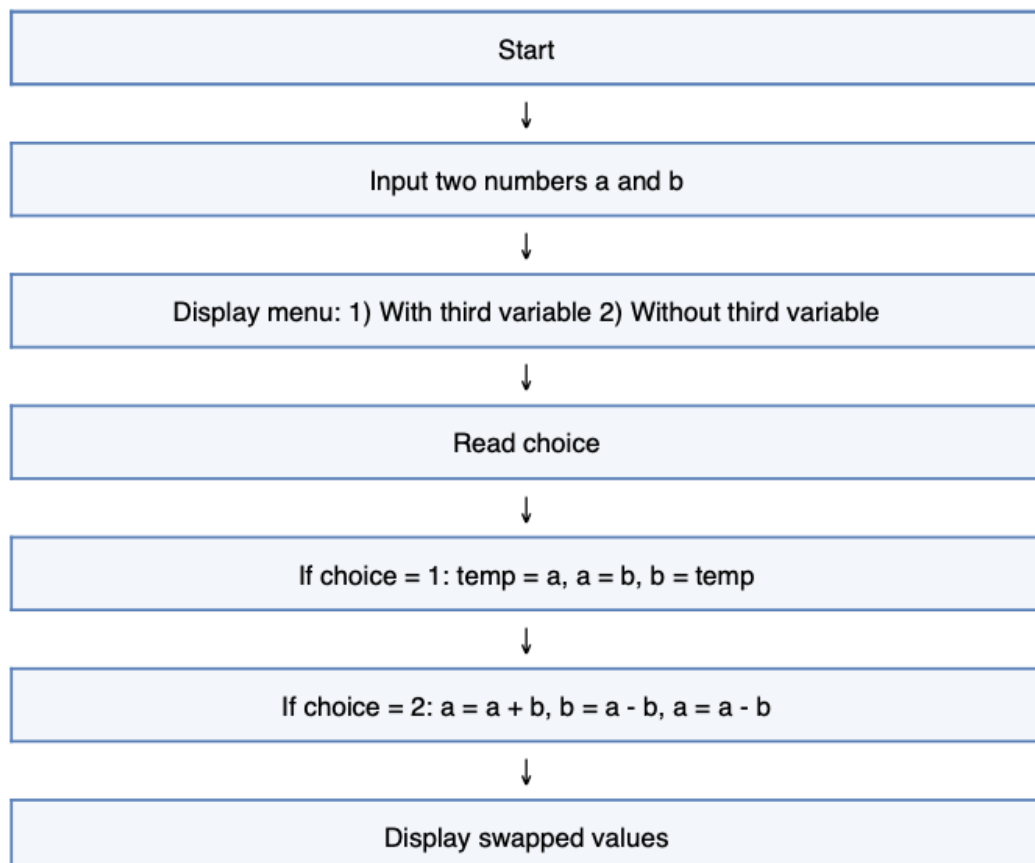
Algorithm

1. Start
2. Input two numbers a and b
3. Display menu: 1) With third variable 2) Without third variable
4. Read choice
5. If choice = 1: temp = a, a = b, b = temp
6. If choice = 2: a = a + b, b = a - b, a = a - b
7. Display swapped values
8. Stop

Pseudocode

```
BEGIN
INPUT a, b
DISPLAY menu options
INPUT choice
IF choice = 1 THEN use temp to swap
ELSE IF choice = 2 THEN swap using arithmetic
ELSE DISPLAY invalid choice
DISPLAY a, b
END
```

Flowchart





Stop

Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	a=12, b=2, choice=1	a=2, b=12	a=2, b=12	Pass
2	a=12, b=2, choice=2	a=2, b=12	a=2, b=12	Pass
3	a=12, b=2, choice=6	Invalid choice	Invalid choice	Pass

Question 1: Swap Two Numbers - Code / Execution Screenshots

Execution screenshot provided:

```
distanceC > ./a.out
printf("Distance from X-axis = %2f\n", distance_x);
printf("Distance from Y-axis = %2f\n", distance_y);
return 0;
}
0
Get Help      WriteOut    Read File    Prev Pg      Cur Pot
to $Exit.    Justify     Where is     Next Pg      To Spell

Last login: Wed Sep  2 17:34:46 on console
Restored session: Tue Sep  3 19:32:44 IST 2026
GurveerSingh@Gurveer-MacBook-Air ~ % nano day9.c
GurveerSingh@Gurveer-MacBook-Air ~ % 
GurveerSingh@Gurveer-MacBook-Air ~ % gcc day9.c
Enter two numbers: 12 ./a.out
1. Swap using third variable
2. Swap without using third variable
Enter your choice: 1
Invalid choice
GurveerSingh@Gurveer-MacBook-Air ~ %
```

Question 2: Sum of Natural Numbers / Sum of Squares

Problem Statement

Write a menu-driven C program to find either the sum of the first n natural numbers or the sum of squares of the first n natural numbers.

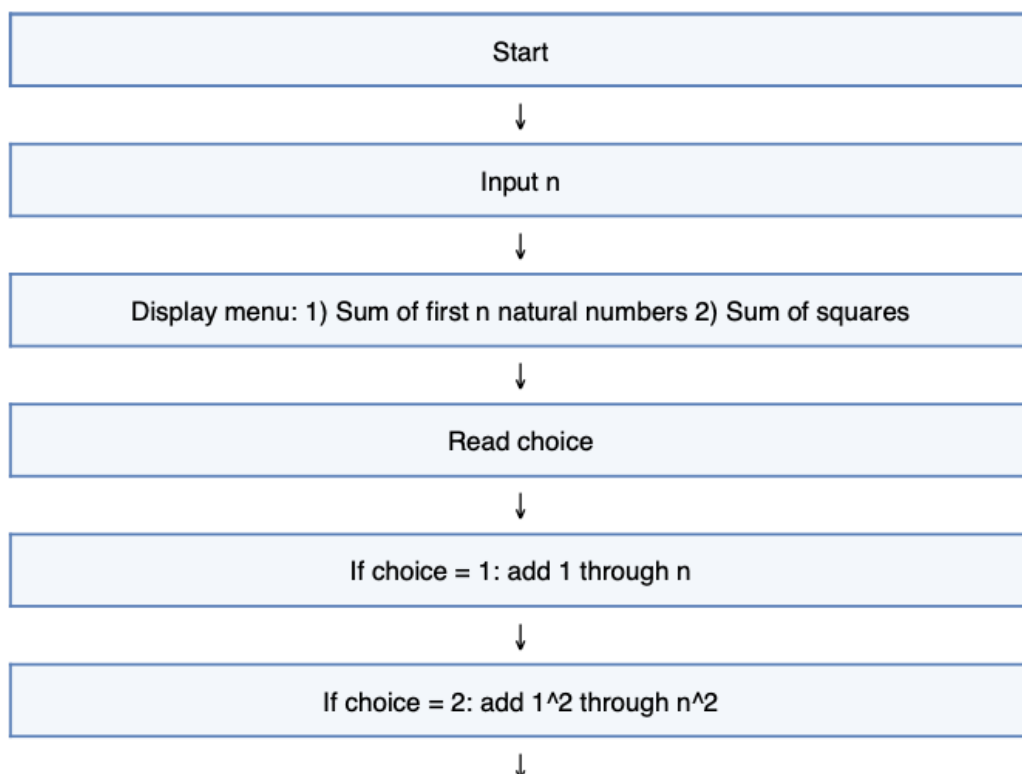
Algorithm

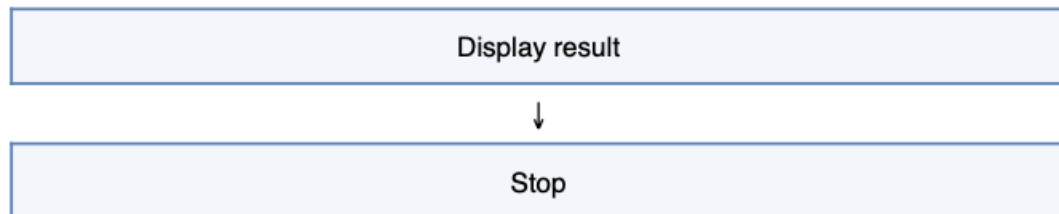
1. Start
2. Input n
3. Display menu: 1) Sum of first n natural numbers 2) Sum of squares
4. Read choice
5. If choice = 1: add 1 through n
6. If choice = 2: add 1^2 through n^2
7. Display result
8. Stop

Pseudocode

```
BEGIN
INPUT n
DISPLAY menu
INPUT choice
SET sum = 0
IF choice = 1 THEN repeat i=1 to n: sum = sum + i
ELSE IF choice = 2 THEN repeat i=1 to n: sum = sum + i*i
ELSE DISPLAY invalid choice
DISPLAY sum
END
```

Flowchart



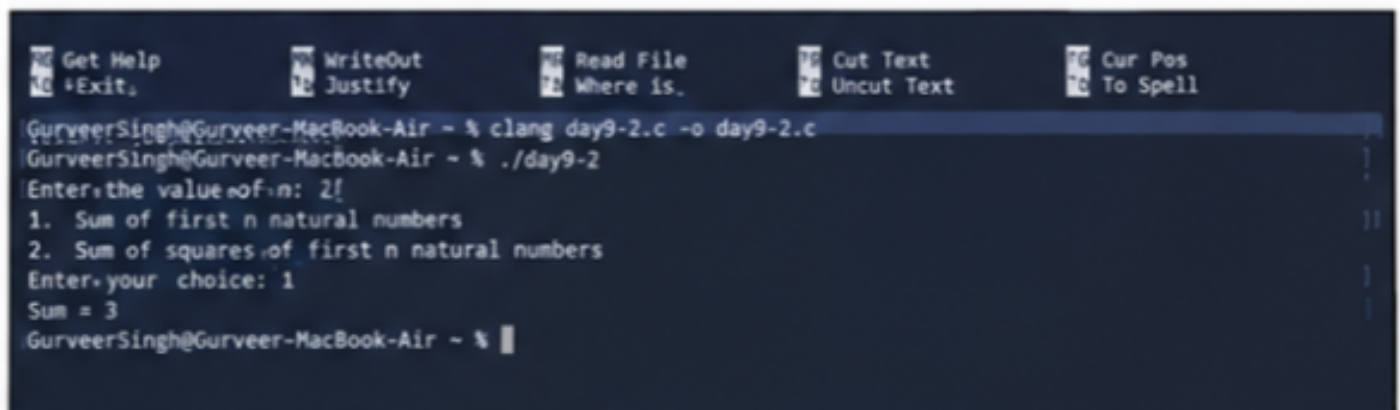


Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	n=2, choice=1	Sum = 3	Sum = 3	Pass
2	n=3, choice=2	Sum of squares = 14	Sum of squares = 14	Pass
3	n=12, choice=12	Invalid choice	Invalid choice	Pass

Question 2: Sum of Natural Numbers / Sum of Squares - Code / Execution Screenshots

Execution screenshots provided:



```
Get Help      WriteOut      Read File     Cut Text      Our Pos
+Exit,        Justify       Where is,     Uncut Text    To Spell

GurveerSingh@Gurveer-MacBook-Air ~ % clang day9-2.c -o day9-2
GurveerSingh@Gurveer-MacBook-Air ~ % ./day9-2
Enter the value of n: 2
1. Sum of first n natural numbers
2. Sum of squares of first n natural numbers
Enter your choice: 1
Sum = 3
GurveerSingh@Gurveer-MacBook-Air ~ %
```

Question 3: Interest Calculations

Problem Statement

Write a menu-driven C program to calculate simple interest, compound amount, compound interest, or total payable amount for a given principal, rate, and time.

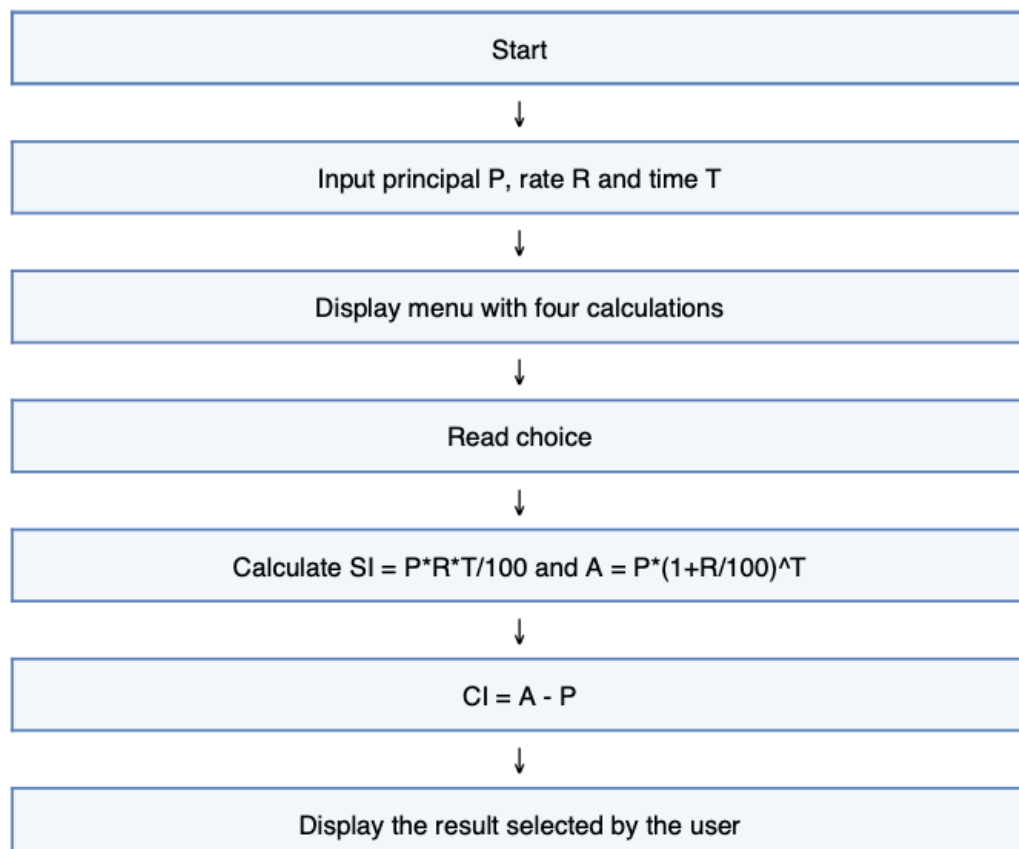
Algorithm

1. Start
2. Input principal P, rate R and time T
3. Display menu with four calculations
4. Read choice
5. Calculate $SI = P \cdot R \cdot T / 100$ and $A = P \cdot (1 + R/100)^T$
6. $CI = A - P$
7. Display the result selected by the user
8. Stop

Pseudocode

```
BEGIN
INPUT P, R, T
DISPLAY 1. Simple Interest, 2. Compound Amount, 3. Compound Interest, 4. Total Payable
INPUT choice
SI = (P*R*T)/100
A = P*(1+R/100)^T
CI = A-P
IF choice=1 DISPLAY SI; choice=2 DISPLAY A; choice=3 DISPLAY CI; choice=4 DISPLAY P+SI
END
```

Flowchart





Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	P=1000, R=10, T=2, choice=1	Simple Interest = 200.00	200.00	Pass
2	P=1000, R=10, T=2, choice=2	Compound Amount = 1210.00	1210.00	Pass
3	P=1000, R=10, T=2, choice=3	Compound Interest = 210.00	210.00	Pass
4	P=1000, R=10, T=2, choice=4	Total Payable Amount = 1200.00	1200.00	Pass

Question 3: Interest Calculations - Code / Execution Screenshots

Code screenshots provided (no typed code included):

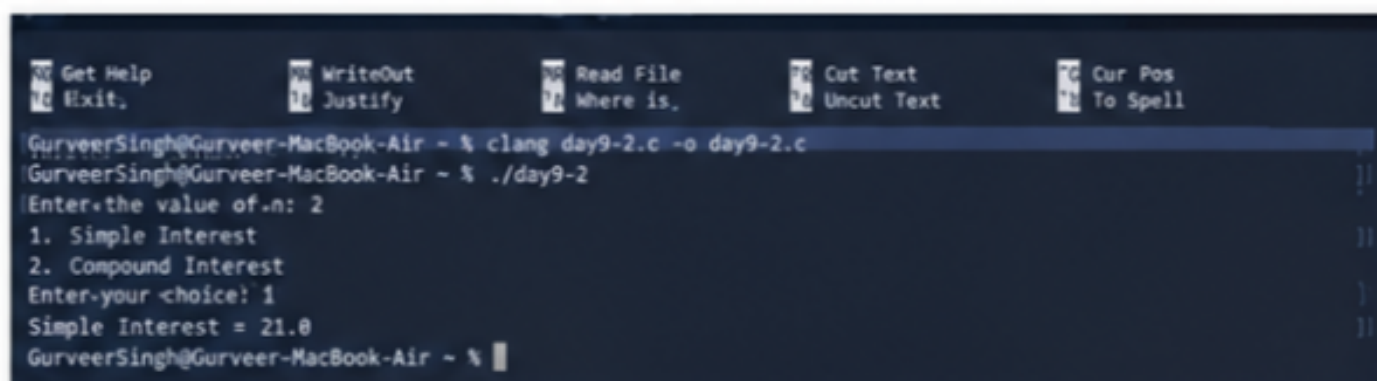
```
Get Help      WriteOut    Read File    Prev Pg      Cut Text     Cur Pos
^C ^X         Justify      ^W Where is. ^P Next Pg    ^C Uncut Text ^O To Spell

GurveerSingh@Gurveer-MacBook-Air ~ % clang day9-2.c -o day9-2.c
GurveerSingh@Gurveer-MacBook-Air ~ % ./day9-2
Enter the value of n: 2
1. Simple Interest
2. Compound Interest
Enter your choice: 1
Simple Interest = 21.0
GurveerSingh@Gurveer-MacBook-Air ~ %
```

```
Get Help      WriteOut    Read File    Prev Pg      Cut Text     Cur Pos
^C ^X         Justify      ^W Where is. ^P Next Pg    ^C Uncut Text ^O To Spell

GurveerSingh@Gurveer-MacBook-Air ~ % clang day9-2.c -o day9-2.c
GurveerSingh@Gurveer-MacBook-Air ~ % ./day9-2
Enter the value of n: 2
1. Simple Interest
2. Compound Interest
Enter your choice: 2
Compound Interest = 5.21
GurveerSingh@Gurveer-MacBook-Air ~ %
```

Question 3: Interest Calculations - Code Screenshot Continued



```
Get Help      WriteOut      Read File     Cut Text      Cur Pos
Exit,         Justify       Where is,     Uncut Text    To Spell

GurveerSingh@Gurveer-MacBook-Air ~ % clang day9-2.c -o day9-2
GurveerSingh@Gurveer-MacBook-Air ~ % ./day9-2
Enter the value of n: 2
1. Simple Interest
2. Compound Interest
Enter your choice: 1
Simple Interest = 21.0
GurveerSingh@Gurveer-MacBook-Air ~ %
```

Conclusion

The Day-9 activities demonstrate the use of menu-driven programming, switch-case selection, arithmetic operations, loops, and mathematical formulas in C. The programs were prepared and tested using representative test cases.